

Functional Dependency

If for every unique value x in column A there exists a value y in column B then B is functionally dependent on A

$$A \rightarrow B$$

A	B	C	D	E
a1	b1	c1	d1	e1
a2	b1	c2	d2	e1
a3	b2	c1	d1	e1
a4	b2	c2	d2	e1
a5	b3	c3	d1	e1

Table R

$$D \rightarrow E$$

$$d1 \rightarrow e1$$

$$d2 \rightarrow e1$$

$$D \not\rightarrow E$$

$$d1 \rightarrow e1$$

$$d1 \rightarrow e3$$

$$C \rightarrow D \quad \checkmark$$

$$D \rightarrow C \quad \times$$

$$A \rightarrow B \quad \checkmark$$

$$B \rightarrow A \quad \times$$

$$\text{Roll\#} \rightarrow \text{Name}$$

$$\times B \rightarrow A$$

$$\times C \rightarrow A$$

$$\checkmark BC \rightarrow A$$

$$f(b, c)$$

$$B \rightarrow B$$

$$BC \rightarrow A \quad \checkmark$$

$$BE \rightarrow A \quad \times$$

Is BC a superkey

$$BC \rightarrow BC$$

$$BC \rightarrow A$$

$$A \rightarrow D \Rightarrow BC \rightarrow D$$

$$A \rightarrow E \Rightarrow BC \rightarrow E$$

$$BC \rightarrow ABCDE$$

w	x	y	z

Armstrong Axioms / Rules

1.

If $B \subseteq A$ then

$$A \rightarrow B$$

$A \rightarrow A$ (trivial case)

$$A = \{x, y, z\}$$

$$B = \{y\}$$

$$f(x, y, z) \quad x$$

Reflexivity

getAge (fname, lname, age, sex)

2.

If $A \rightarrow B$ then

$$AX \rightarrow BX$$

Augmentation

}

set

fname $\rightarrow$ sex

then

fname, age $\rightarrow$ sex, age

3.

If $A \rightarrow B$ and $B \rightarrow C$

then $A \rightarrow C$

Transitivity

{ For every x there
is a $y > x$

$$x+1 = x$$

$$\Rightarrow 2+2 = 2$$

4.

If $A \rightarrow B$ & $A \rightarrow C$

then $A \rightarrow BC$

Union

$$\boxed{x+1 > x}$$

5.

If $A \rightarrow B$ & $x \rightarrow y$

then $AX \rightarrow BY$

Composition

$$f(a) \rightarrow b$$

$$g(b) \rightarrow c$$

$$g(f(a)) \rightarrow c$$

6.

If $A \rightarrow BC$ then

$A \rightarrow B$ & $A \rightarrow C$

(Inverse of union)

Decomposition

$$AX \rightarrow BY$$

7.

If $A \rightarrow B$ & $BC \rightarrow D$

then $AC \rightarrow D$

Pseudo-Transitivity

$$\left[\begin{array}{l} A \rightarrow B \\ x \rightarrow y \end{array} \right] \times$$

$$\left[\begin{array}{l} AX \rightarrow B \\ AX \rightarrow y \end{array} \right] \checkmark$$

$R(A, B, C, D, E, F)$

$A \rightarrow B \quad A \rightarrow C, \quad C \rightarrow D, \quad E \rightarrow F$

$A \rightarrow A$ reflexivity

$A \rightarrow B$ given

$A \rightarrow C$ given

$A \rightarrow D$ transitively

$A \rightarrow ABCD$ By union

$AE \rightarrow ABCDEF$

By composition

can be reversed

decomposition

cannot be reversed

$BD \rightarrow A$

$BC \rightarrow E$

$A \rightarrow F$

$BD \rightarrow AF$

$BCD \rightarrow AEF$

Given then

We can determine

$BD \rightarrow A$	$BC \rightarrow E$	$A \rightarrow F$
$\underline{BCD} \rightarrow A$	$\underline{BCD} \rightarrow E$	$\underline{BCD} \rightarrow A \rightarrow F$

$BCD \rightarrow AEF$

$\{B, D\} \cup \{B, C\} \rightarrow \{A\} \cup \{E\} \cup \{F\}$

$\{B, C, D\} \rightarrow \{A, E, F\}$

$ABC \rightarrow XYZ$

$\{A, B, C\} \rightarrow \{X, Y, Z\}$

$\{X, Y, Z\} \rightarrow \{X\}$

Super key

Any set of attributes which can derive ^{values for} all other attributes

$A \rightarrow A, B, C, D, E, F$
 $\uparrow$ set of attributes
 $\underbrace{A, B, C, D, E, F}$ all attributes in relation

Closure

$R(A, B, C, D, E)$

$A \rightarrow BC, C \rightarrow B, D \rightarrow E, E \rightarrow D$

$\{A\}^+ \rightarrow ABC$

$B^+ \rightarrow B$

$C^+ \rightarrow BC$

$D^+ \rightarrow DE$

$E^+ \rightarrow DE$

$AB^+ \rightarrow ABC$

$AC^+ \rightarrow ABC$

$AD^+ \rightarrow ABCDE$ ✓ superkey

$AE^+ \rightarrow ABCDE$ ✓ superkey

$ABCD^+ \rightarrow ABCDE$ ✓ superkey

$ABCDE^+ \rightarrow ABCDE$ ✓ superkey

$ACD^+ \rightarrow ABCDE$ } superkey
 $\quad \quad \quad \rightarrow ACDBE$ }

$R(A, B, C, D, E)$

$A \rightarrow B, AC \rightarrow D, D \rightarrow C$

$A^+ \rightarrow AB$

$AB \rightarrow AB$

$BD \rightarrow BCD$

$B^+ \rightarrow B$

$AC \rightarrow ABCD$

$BE \rightarrow BE$

$C^+ \rightarrow C$

$AD \rightarrow ABCD$

$CD \rightarrow CD$

$D^+ \rightarrow CD$

$AE \rightarrow ABE$

$CE \rightarrow CE$

$E^+ \rightarrow E$

$BC \rightarrow BC$

$DE \rightarrow CDE$

ABC

3 ABC

AC BC

AC ABC

If n elements/attributes in relation

then # of possible closures = $2^n - 1$

(7)

$$P(\{A, B, C\}) = \{ \phi, \{A\}, \{B\}, \{C\}, \{A, B\}, \{A, C\}, \{B, C\}, \{A, B, C\} \}$$

2^n

W XYZ

W X Y Z

W X W Y W Z

X Y X Z Y Z

W X Y W X Z

X Y Z W Y Z

W X Y Z (15)

$R(A, B, C, D, E)$

① $A \rightarrow BC$ $C \rightarrow B$ $D \rightarrow E$ $E \rightarrow D$

② $A \rightarrow B$ $B \rightarrow C$ $D \rightarrow C$

25-1-31

HW

Find all possible closures
for the above relations
with given functional dependencies

Also identify the superkey

1	A^+
2	B^+
3	C^+
4	D^+
5	E^+
6	AB^+
7	AC^+
8	AD^+
9	AE^+
10	BC^+
11	BD^+
12	BE^+
13	CD^+
14	CE^+
15	DE^+
16	ABC^+
17	ABD^+
18	ABE^+
19	ACD^+
20	ACE^+
21	ADE^+
22	BCD^+
23	BCE^+
24	BDE^+
25	CDE^+
26	$ABCD^+$
27	$ABCE^+$
28	$ABDE^+$
29	$ACDE^+$
30	$BCDE^+$
31	$ABCDE^+$

Candidate Key : All superkeys which have the minimum # of attributes involved

Superkey = { AD, AE, ABD, ACD, ABE, ADE, ABCD, ABCE, ABDE }
 Candidate key = { AD, AE }

R(A, B, C, D, E) $AB \rightarrow C$ $B \rightarrow D$ $C \rightarrow E$ $D \rightarrow A$

$A^+ \rightarrow A$

$B^+ \rightarrow$ $\checkmark$

Candidate key = { B }

$C^+ \rightarrow CE$

$D^+ \rightarrow DA$

$E^+ \rightarrow E$

AB^+

R(w, x, y, z)

$z \rightarrow wxy$

$x \rightarrow w$

$y \rightarrow z$

L	R
	w
	x
	y
	z

$w^+ \rightarrow w$

$w^+ \rightarrow w$

$x^+ \rightarrow xw$

$x^+ \rightarrow xw$

$y^+ \rightarrow y$

$y^+ \rightarrow yzwxy \checkmark$

$z^+ \rightarrow zwxxy \checkmark$

$z^+ \rightarrow zwxxy \checkmark$

L	B	R
	y	w
	z	
	x	

C.K. = { z }

CK = { y, z }

x^+
y^+
z^+
x, y^+
x, z^+
y, z^+
x, y, z^+

R(ABCDE)

$A \rightarrow C$

$C \rightarrow B$

$B \rightarrow D$

$D \rightarrow E$

$A^+ \rightarrow ACBDE \checkmark$

$B^+ \rightarrow BDE$

C.K. = { A }

$C^+ \rightarrow CBDE$

$D^+ \rightarrow DE$

$E^+ \rightarrow E$

L	R
<u>A</u>	C
	E
	B
	D

L	B	R
A	C	E
	B	
	D	

$A^+ \rightarrow$
 AB
 AC
 AD

ABC
 ABD

$R(ABCDE)$

$A \rightarrow C \quad E \rightarrow D \quad B \rightarrow C$

$A^+ \rightarrow AC$

$B^+ \rightarrow BC$

$C^+ \rightarrow C$

$D^+ \rightarrow D$

$E^+ \rightarrow ED$

L	R
A	C
B	D
E	

L	B	R
A		C
B		D
C		

$ABC^+ \rightarrow$

AB^+

AC^+

AD^+

AE^+

BC^+

BD^+

BE^+

CD^+

CE^+

DE^+

ABC^+

ABD^+

$ABE^+ \rightarrow ABEC D \checkmark$

BCD

BCE

$\vdots$

$CK = \{ABE\}$

HW #2

Find the Candidate keys in the following relations & FD pairs

① $AB \rightarrow C$ $BC \rightarrow D$ $CD \rightarrow A$ $R(ABCD)$

② $A \rightarrow BCD$ $C \rightarrow A$ $R(ABCD)$

③ $A \rightarrow B$ $BC \rightarrow E$ $ED \rightarrow A$ $R(ABCDE)$

④ $AB \rightarrow C$ $DE \rightarrow B$ $CD \rightarrow E$ "

⑤ $AB \rightarrow C$ $C \rightarrow D$ $B \rightarrow EA$ "

⑥ $A \rightarrow B$ $BC \rightarrow E$ $ED \rightarrow A$ "

⑦ $C \rightarrow D$ $C \rightarrow A$ $B \rightarrow C$ "

⑧ $B \rightarrow C$ $D \rightarrow A$ "

⑨ $ABC \rightarrow D$ $D \rightarrow A$ "

⑩ $A \rightarrow B$ $BC \rightarrow D$ $A \rightarrow C$ $R(ABCD)$

⑪ $AB \rightarrow C$ $AD \rightarrow D$ $C \rightarrow A$ $D \rightarrow B$
 $R(ABCDEFGH)$

⑫ $AB \rightarrow C$ $AC \rightarrow B$ $AD \rightarrow E$ $B \rightarrow D$ $BC \rightarrow A$ $E \rightarrow G$

⑬ $A \rightarrow C$ $C \rightarrow A$ $B \rightarrow D$ $AB \rightarrow E$ $BC \rightarrow E$ $R(ABCDE)$

⑭ $AB \rightarrow EF$ $A \rightarrow C$ $B \rightarrow DGH$ $H \rightarrow I$ $AE \rightarrow BDFGHI$
 $R(ABCDEFGH)$

⑮ $A \rightarrow BE$ $C \rightarrow BE$ $C \rightarrow D$ $E \rightarrow F$ $R(ABCDEF)$

$R(A B C D)$

$AB \rightarrow C$

$BC \rightarrow D$

$CD \rightarrow A$

L	B	R
B	A	
	C	
	D	

$B^+ \rightarrow x$

$AB^+ \rightarrow x$

$BC^+ \rightarrow x$

$BD^+ \rightarrow x$

$B^+ \rightarrow B$

$BA^+ \rightarrow ABCD$

$BC^+ \rightarrow ABCD$

$BD^+ \rightarrow BD$

$CK = \{ AB, BC \}$

$BA^+ \rightarrow x$

$BAD^+ \rightarrow x$

$BCD^+ \rightarrow x$

$BCAD$

12 $R(A B C D E F G H)$

$AB \rightarrow C$

$AC \rightarrow B$

$AD \rightarrow E$

$B \rightarrow D$

$BC \rightarrow A$

$E \rightarrow G$

L	B	R
F	A	G
H	B	
	C	
	D	
	E	

$FH^+ \rightarrow FH$

$FHA^+ \rightarrow FHA$

$FHB^+ \rightarrow FHB$

$FHC^+ \rightarrow FHC$

$FHD \rightarrow FHD$

$FHE \rightarrow FHEG$

$FHAB \rightarrow FHAB CDEG$

$FHAC \rightarrow FHACD$

$FHAD \rightarrow FHAD E$

$FHAE$

$FHBC \rightarrow$

$FHBD$

$FHBE$

FH

Database Design Guidelines

1. Table should have relevant data
2. Avoid redundancy — anomalies
 - insert
 - update
 - delete
3. Avoid using NULL values
4. Avoid generation of spurious records (using Natural Join)

Normalization

0. 0 - Normal Form

No column should have multiple values in any row

1. 1st Normal Form

Data must be in 0-NF and table should have a PK.

2. 2nd Normal Form

Table must be in 1st NF and

All Non key Attributes should be fully functionally dependent on the complete primary key
OR

There should be no partial dependency b/w non key attributes & the primary key

3. 3rd Normal Form

Table must be in 2nd NF and

Non key attributes must be directly dependent on the PK
OR

There should be no transitive dependency in the table

item	colors	price	tax	qty
T-shirt	red, blue	12.00	0.60	5,3
polo	red, yellow	15.00	0.60	8,2
T-shirt	red, black	12.00	0.60	5,9
sweatshirt	blue, black	25.00	1.25	12,2

0 - Normal Form

	<u>item</u>	<u>color</u>	<u>price</u>	<u>tax</u>	<u>qty</u>	<u>F.D</u>
item → price ✓	T-shirt	red	12	0.60	5	item → price
color → tax X	T-shirt	blue	12	0.60	3	price → tax
color → qty X	polo	red	15	0.60	8-3=5	item, color → qty
price → tax ✓	polo	yellow	15	0.60	2	
price → item X	T-shirt	black	12	0.60	9	
color, qty → item ?	S. shirt	blue	25	1.25	12	
color, item → qty ✓	S. shirt	black	25	1.25	2	

1st Normal Form

$(item, color)^+ \rightarrow item, color, price, tax, qty$

L	B	R
item	price	tax
color		qty

$Ch = \{(item, color)\}$

Primary key = item, color

2nd Normal Form

$R(\underline{item}, \underline{color}, price, tax, \boxed{qty})$

item, color

$item^+ \rightarrow item, price, tax$

$color^+ \rightarrow color$

R2

<u>item</u>	price	tax
T-shirt	12	0.60
polo	15	0.60
S-shirt	25	1.25

R3

<u>item</u>	<u>color</u>	<u>qty</u>
T-shirt	red	5
T-shirt	blue	3
T-shirt	black	9
polo	red	8
polo	yellow	2
S-shirt	blue	12
S-shirt	black	2

R2 & R3 are in 2nd normal form

$R2(\underline{\text{item}}, \text{price}, \text{tax})$
 $R3(\underline{\text{item}}, \underline{\text{color}}, \text{qty})$

$R4(\underline{\text{item}}, \text{price})$
 $R5(\underline{\text{price}}, \text{tax})$
 $R3(\underline{\text{item}}, \underline{\text{color}}, \text{qty})$

3rd NF

R4

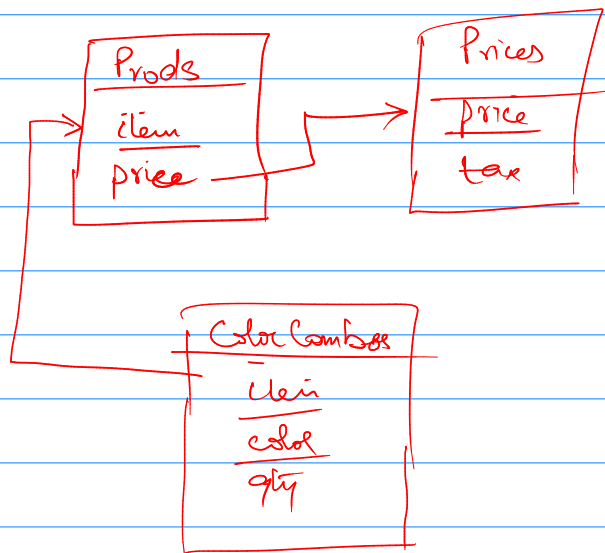
<u>item</u>	price
T-shirt	12
Polo	15
S-shirt	25

R5

<u>price</u>	tax
12	0.60
15	0.60
25	1.25

R3

<u>item</u>	<u>color</u>	qty
T-shirt	red	5
T-shirt	blue	3
T-shirt	black	9
Polo	red	8
Polo	yellow	2
S-shirt	blue	12
S-shirt	black	2



R(A B C D K M N O S T)

$A \rightarrow M$, $C \rightarrow NO$, $AB \rightarrow K$ $BC \rightarrow S$
 $S \rightarrow T$ $ABC \rightarrow D$ $D \rightarrow E$

Normalize to 3rd NF

1st NF find a PK

L	B	R
A	S	D E
B	D	K
C		M N O T

$(ABC)^+ \rightarrow \underline{A} \underline{B} \underline{C} \underline{M} \underline{N} \underline{O} \underline{K} \underline{S} \underline{T} \underline{D} \underline{E}$

CK = { ABC }

R1 (A B C ~~D~~ E K M N O S T)

R1 is in 1NF

2NF No partial dependency b/w non prime attributes & part of the PK

PK = { ABC }

NPrime = { D ~~M~~ ~~N~~ ~~O~~ ~~S~~ ~~T~~ ~~E~~ }

$A^+ \rightarrow AM$

$B^+ \rightarrow B$

$C^+ \rightarrow CNO$

$AB^+ \rightarrow ABK$ ~~M~~

$AC^+ \rightarrow AC$ ~~M~~ ~~N~~ ~~O~~

$BC^+ \rightarrow BC$ ~~S~~ ~~T~~

$R(\underline{AB} M K)$
$A^+ \rightarrow M$ $R(\underline{A} M)$
$B^+ \rightarrow B$ $T(\underline{AB} K)$

R2 (A M)

R3 (C N O)

R4 (A B K)

R5 (B C S T)

R6 (A B C ~~D~~) ~~E~~

2NF

3rd NF : No transitive dependency b/w non prime attributes

R2 (A → M) ✓

R3 (C → N → O) ✓

R4 (A → B → K) ✓

R5 (B → S → T)

R7 (B → S) R8 (S → T)

R6 (A → B → C → D) E

R9 (A → B → C → D) R10 (D → E)

R2, R3, R4, R7, R8, R9, R10 are in 3NF

A → B	S → T
BC → D	ST → U
AC → D	SS → U
	S → U

HW#3 Due on Wednesday before class

For each of the following, find relevant FD and resolve till 3NF

Full Names	Physical Address	Movies rented	Salutation	Category
Janet Jones	First Street Plot No 4	Pirates of the Caribbean,	Ms.	Action, Action
"	"	Clash of the Titans	Ms.	Action
Robert Phil	3 rd Street 34	Forgetting Sarah Marshal,	Mr.	Romance, Romance
		Daddy's Little Girls		Romance
Robert Phil	5 th Avenue	Clash of the Titans	Mr.	Action

$R_1(\underline{FN}, \underline{PA}, M, S, C)$
 $R_2(\underline{FN}, S)$ $R_3(\underline{M}, C)$ $R_4(\underline{FN}, \underline{PA}, \underline{M})$
 $\xrightarrow{R_2, R_3 \& R_4}$ 2NF & 3NF $\rightarrow$ 1NF

Project Management Report				
Project Code: PC010		Project Title: Pensions System		
Project Manager: M Phillips		Project Budget: £24,500		
Employee No.	Employee Name	Department No.	Department Name	Hourly Rate
S10001	A Smith	L004	IT	£22.00
S10030	L Jones	L023	Pensions	£18.50
S21010	P Lewis	L004	IT	£21.00
S00232	R Smith	L003	Programming	£26.00
Total Staff on Project: 4			Average Hourly Rate:	£21.88

$PC \rightarrow PT, PM, PB$ $D\# \rightarrow DN$
 $E\# \rightarrow EN, D\#$ $PC, E\# \rightarrow HR$

SALESPERSON NUMBER	SALESPERSON NAME	SALES AREA	CUSTOMER NUMBER	CUSTOMER NAME	WAREHOUSE NUMBER	WAREHOUSE LOCATION	SALES AMOUNT
3462	Waters	West	18765	Delta Systems	4	Fargo	13540
			18830	A. Levy and Sons	3	Bismarck	10600
			19242	Ranier Company	3	Bismarck	9700
3593	Dryne	East	18841	R. W. Flood Inc.	2	Superior	11560
			18899	Seward Systems	2	Superior	2590
			19565	Stodola's Inc.	1	Plymouth	8800
etc.							

$SP\# \rightarrow SN, SA$
 $C\# \rightarrow CN$
 $W\# \rightarrow WL$

$SP\#, C\#, W\# \rightarrow$ Sales Amount

Course Code	23	Department	Business
Course Title	Business Accounting	Tutor	Trevor Albright
Level	3	Dept Head	Mr Alan Parkes
Mode of Attendance	Part Time		
Course Fee	£55.00		
Start Date	20/08/2016		
End Date	24/08/2016		

Sid, Ccode → E date, Enrolled By

Student ID Num.	Student Name	Student Address	Student Telephone	Student DOB	Enrolment Date	Enrolled By:-
14111	Alan Briggs	24 Abbey Park, Bangor, BT19 1RT	028 91468665 07798754623	10/08/94	10/06/16	Andrea Gilchrist
14332	Susan McBride	19 Belmont Road, Bangor, BT19 3TR	028 91488930 07891454235	01/03/93	03/07/16	David Smythe
14378	David Adair	14 Grange Park, Bangor, BT19 4TF	028 91584630 07891834277	06/11/87	05/07/16	Andrea Gilchrist

CCode → CTitle, Level, Attendance, Fee, Start, End, Dept, Tutor

Dept → DName

Sid → SName, SAddress, SDOB,

Sid, Code → Edate, Enrolled By

Title	<u>Author1</u>	<u>Author2</u>	ISBN	Subject	Pages	Publisher
Database System Concepts	Abraham Silberschatz	Henry F. Korth	0072958863	MySQL, Computers	1168	McGraw-Hill
Operating System Concepts	Abraham Silberschatz	Henry F. Korth	0471694665	Computers	944	McGraw-Hill

ISBN → Title, Pages, Publisher

CCode $\rightarrow$ CTitle, level, Attendance, Fee, Start, End, Dept, Tutor

Dept $\rightarrow$ DName

Sid $\rightarrow$ SName, SAddress, SDOB,

Sid, Code $\rightarrow$ EDate, Enrolled By

First Normal Form

PK (Ccode, Sid, Telephone)

L	B	R
Ccode	Dept	CTitle Fee
Sid		level SDate EDate
STelephone		Attend Tutor
		SAdd SName, SDOB

2nd NF

R1 (Ccode, CTitle, level, Fee, Attend, SDate, EDate, Tutor, Dept, DName)

R2 (Sid, SName, SDOB, SAddress)

R3 (Sid, Ccode, STelephone)

3rd NF

R2 is ok

R3 is ok

R1 is split

R4 (Dept, DName)

R5 (Ccode, CTitle, level, Fee, Attend, SDate, EDate, Tutor, Dept)

$A \rightarrow C\# \rightarrow C\overset{B}{Name}, C\overset{C}{Phone}, C\overset{D}{Address}$
 $E\# \rightarrow E\overset{F}{Name}, E\overset{G}{Phone}, E\overset{H}{Address}, E\overset{I}{Sal}$
 $P\# \rightarrow P\overset{K}{Name}, P\overset{L}{Desc}, P\overset{M}{Price}$
 $O\# \rightarrow O\overset{O}{Date}, O\overset{E}{\#}, O\overset{A}{C\#}$
 $O\#, P\# \rightarrow q\overset{P}{ly}$

$A \rightarrow BCD$
 $E \rightarrow FGHI$
 $J \rightarrow KLM$
 $N \rightarrow OEA$
 $NI \rightarrow P$

$(JN)^+ \rightarrow J N P O A E B C D F G H I E$
 $K L M$

L	B	R
J	A	B
N	E	C
		D
		E
		F
		G
		H
		I
		J
		K
		L
		M
		N
		O
		P

$CK = \{JN\}$

1st NF

$R(A B C D E F G H I J K L M N O P)$

$J^+ \rightarrow J K L M$

$R_2(\underline{J} K L M)$

$N^+ \rightarrow N O E A B C D F G H I$

$R_3(A B C D E F G H I \underline{N} O)$

$R_4(\underline{J} \underline{N} P)$

2NF

$R_2(\underline{J} K L M)$

$R_3(A B C D E F G H I \underline{N} O)$

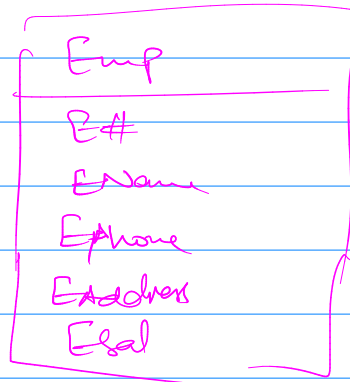
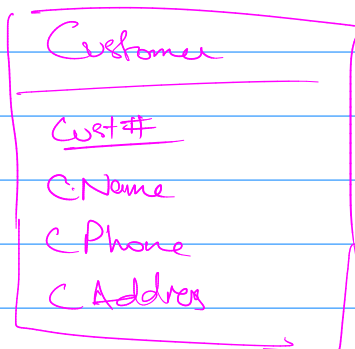
$R_5(\underline{N} E A O)$

$R_6(\underline{A} B C D)$

$R_7(\underline{E} F G H I)$

$+R_2 \& R_4$

3NF



$C\# \rightarrow CName, CPhone, CAddress$
 $E\# \rightarrow EName, EPhone, EAddress, ESsal$
 $P\# \rightarrow PName, PDesc, Pprice$
 $O\# \rightarrow ODate, E\#, C\#$
 $O\#, P\# \rightarrow qty$

$R_4(I, N, P)$

$R_2(I, K, L, M)$

$R_5(N, E, A, O)$

$R_6(A, B, C, D)$

$R_7(E, F, G, H, I)$

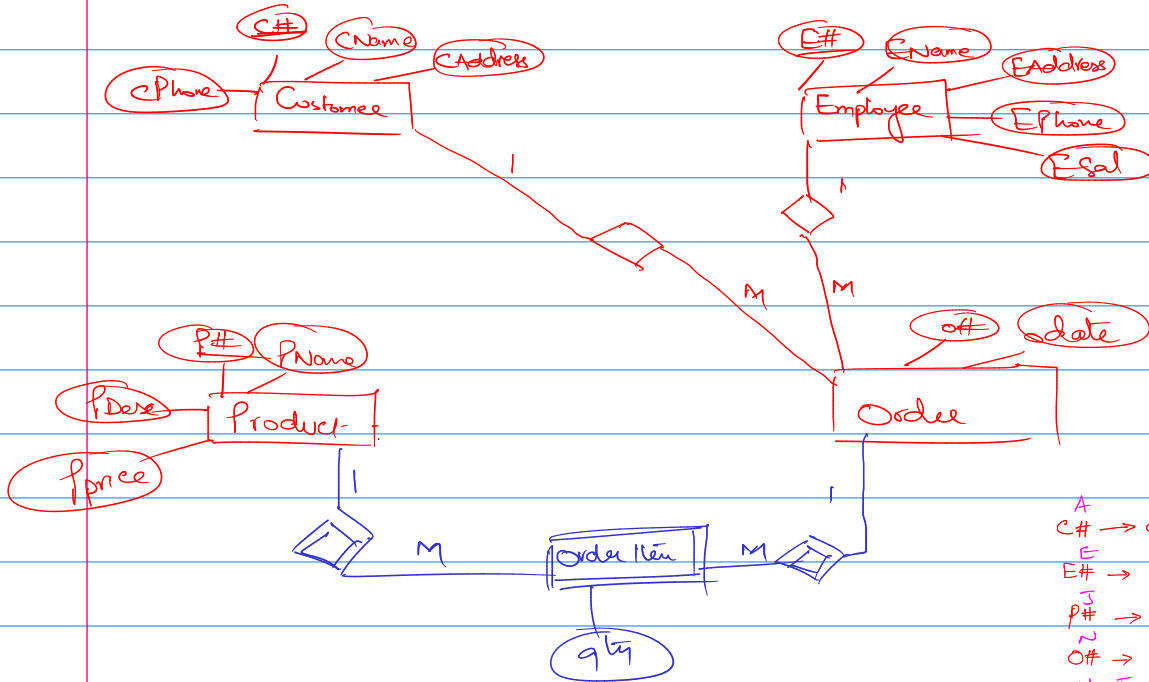
R2	
<u>I</u>	I#
K	PName
L	PDesc
M	PPrice

R4	
<u>I</u>	P#
N	O#
P	qty

R5	
<u>N</u>	O#
A	C#
E	E#
O	odate

R6	
A	C#
B	CName
C	CPhone
D	CAddress

R7	
<u>E</u>	E#
F	EName
G	EPhone
H	EAddress
I	ESsal



$C\# \rightarrow CName, CPhone, CAddress$
 $E\# \rightarrow EName, EPhone, EAddress, ESsal$
 $P\# \rightarrow PName, PDesc, PPrice$
 $O\# \rightarrow ODate, E\#, C\#$
 $O\#, P\# \rightarrow qty$

Cust	
A	
B	
C	
D	

Customer	
<u>C#</u>	A
CName	B
CAddress	D
CPhone	C

Emp	
<u>E#</u>	E
EName	F
EAddress	H
EPhone	G
ESsal	I

Emp	
<u>E</u>	
F	
G	
H	
I	

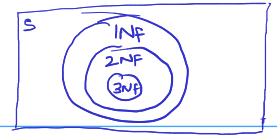
Prod	
<u>P#</u>	J
PName	K
PDesc	L
PPrice	M

Order Items	
<u>O#</u>	N
P#	J
qty	P

Order	
<u>O#</u>	N
C#	A
E#	E
odate	O



Identification of Normal Form



$R(A B C D E F G H)$

ABF

$AB \rightarrow C$

$A \rightarrow DE$

$B \rightarrow F$

$CK = \{AB\}$

$F \rightarrow GH$

S: Sample Space of all possible relations without multivalued attributes

3NF

$\alpha \rightarrow \beta$

α is a CK / SK CR

β is a prime attribute

2NF

$\alpha \rightarrow \beta$

α is not part of CK AND

β is a non-prime attribute

prime Attributes = $\{A, B\}$

3NF 2NF

1NF

$\alpha \rightarrow \beta$

$AB \rightarrow C$

$A \rightarrow D$

$A \rightarrow E$

$B \rightarrow F$

$\rightarrow F \rightarrow G$

$F \rightarrow H$

X or X = X

✓ AND ✓ = ✓

X

1NF

$R(ABCDE)$

$BC \rightarrow ADE$

$D \rightarrow B$

$CK = \{BC, CD\}$

prime attributes = $\{B, C, D\}$

3NF

$BC \rightarrow A$

$BC \rightarrow D$

$BC \rightarrow E$

$D \rightarrow B$

3NF

$R(A\ B\ C\ D\ E)$

$CE \rightarrow D$

$CK = \{CE\}$

$D \rightarrow B$

$C \rightarrow A$

	3NF	2NF	
$CE \rightarrow D$	✓	✓ AND ✓ = ✓	
$D \rightarrow B$	X	✓ AND ✓ = ✓	
$C \rightarrow A$		X AND ? = X	<u>INF</u>
		X	

$R(A\ B\ C\ D\ E)$

$AB \rightarrow CD$

$D \rightarrow A$

$CK = \{AB, BC, BD\}$

prime attributes = $\{A, B, C, D\}$

$BC \rightarrow DE$

	3NF
<u>AB</u> $\rightarrow C$	✓
<u>AB</u> $\rightarrow D$	✓
$D \rightarrow A$	✓
<u>BC</u> $\rightarrow D$	✓
<u>BC</u> $\rightarrow E$	✓
	<u>3NF</u>

HW #4 Identify the normal form in the following relations

$R(VWXY)$

$X \rightarrow VY$

$Y \rightarrow Z$

$Z \rightarrow Y$

$VW \rightarrow X$

$R(ABCDEFGHI I)$

$AB \rightarrow C$

$BD \rightarrow EF$

$AD \rightarrow GH$

$A \rightarrow I$

$R(ABCDEF)$

$ABC \rightarrow D$

$ABD \rightarrow E$

$CD \rightarrow F$

$CDF \rightarrow B$

$BF \rightarrow D$

$R(ABC)$

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow A$

Minimal Cover

1. Only 1 attribute on RHS of every FD

$$AB \rightarrow CD$$

$\Downarrow$

$$AB \rightarrow C$$

$$AB \rightarrow D$$

2. No Extra attribute on LHS

$$\begin{array}{l} A \rightarrow B \\ \underline{AB} \rightarrow C \end{array} \quad \left. \vphantom{\begin{array}{l} A \rightarrow B \\ \underline{AB} \rightarrow C \end{array}} \right\} \begin{array}{l} A \rightarrow B \\ A \rightarrow C \end{array}$$

3. No Redundant FD

$$\begin{array}{c} A \rightarrow B \\ \times \end{array}$$

$$\underline{B \rightarrow C}$$

$$\underline{A \rightarrow C}$$

$$\underline{A \rightarrow B}$$

$$\underline{B \rightarrow C}$$

$$\checkmark \text{ roll \#} \rightarrow \text{name}$$

$\text{name} \rightarrow \text{sex}$

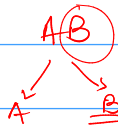
$$\checkmark \text{ roll \#} \rightarrow \text{sex}$$

$$F = \{A \rightarrow BC, B \rightarrow C, AB \rightarrow A\}$$

Step 1

$$F' = \{A \rightarrow B, A \rightarrow C, B \rightarrow C, \underline{AB \rightarrow A}\}$$

Step 2



So B is extra attribute

$$A^+ \rightarrow ABC$$

$$B^+ \rightarrow BC$$

$$F' = \{A \rightarrow B, A \rightarrow C, B \rightarrow C, \underline{A \rightarrow A}\}$$

now $A \rightarrow A$ is trivial and can be removed

$$F' = \{A \rightarrow B, \underline{A \rightarrow C}, B \rightarrow C\}$$

Step 3

$$\underline{A \rightarrow B} \quad A^+ - \{A \rightarrow \underline{B}\} = AC \quad \text{so } A \rightarrow B \text{ is required}$$

$$\underline{A \rightarrow C} \quad A^+ - \{A \rightarrow C\} = \underset{\substack{\uparrow \\ \text{RHS}}}{ABC} \quad \text{so } A \rightarrow C \text{ is redundant}$$

$$F' = \{A \rightarrow B, B \rightarrow C\}$$

$$B \rightarrow C \quad B^+ - \{B \rightarrow C\} = B \quad \text{so } B \rightarrow C \text{ is required}$$

$$F' = \{A \rightarrow B, B \rightarrow C\}$$

$$F = \{A \rightarrow BC, B \rightarrow C, AB \rightarrow A\}$$

$$F = \{ A \rightarrow AB \quad B \rightarrow C \quad A \rightarrow B \quad AB \rightarrow C \}$$

Step 1

$$F' = \{ \underbrace{A \rightarrow A}_{\text{trivial}}, \underbrace{A \rightarrow B}_{\text{duplicate}}, B \rightarrow C, \underbrace{A \rightarrow B}_{\text{duplicate}}, AB \rightarrow C \}$$

$$F' = \{ A \rightarrow B, B \rightarrow C, AB \rightarrow C \}$$

Step 2

$$\begin{array}{c} AB \\ \swarrow \quad \searrow \\ A^+ \quad B \end{array}$$

$$A^+ \rightarrow ABC = \text{don't need } B$$

$$F' = \{ A \rightarrow B, B \rightarrow C, A \rightarrow C \}$$

Step 3

$$A \rightarrow B$$

$$A^+ - \{ A \rightarrow B \} = AC \quad \text{required}$$

$$B \rightarrow C$$

$$B^+ - \{ B \rightarrow C \} = B \quad \text{required}$$

$$A \rightarrow C$$

$$A^+ - \{ A \rightarrow C \} = ABC \quad \text{so redundant}$$

$$F' = \{ A \rightarrow B, B \rightarrow C \}$$

$$AB \rightarrow C$$

$$(AB)^+ \rightarrow \underline{ABC}$$

$$F = \{A \rightarrow D \quad BC \rightarrow AD \quad C \rightarrow B \quad E \rightarrow AD\}$$

Step 1

$$F' = \{A \rightarrow D \quad BC \rightarrow A \quad BC \rightarrow D \quad C \rightarrow B \quad E \rightarrow A, E \rightarrow D\}$$

$$E^+ \rightarrow EAD$$

$$F^+ \rightarrow F$$

Step 2

$$BC$$

$$B^+ \rightarrow B$$

$$C^+ \rightarrow CB$$

B is extra

$$F' = \{A \rightarrow D \quad \underline{C \rightarrow A} \quad \underline{C \rightarrow D} \quad C \rightarrow B \quad E \rightarrow A \quad E \rightarrow D\}$$

Step 3

$$A \rightarrow D \quad A^+ - \{A \rightarrow D\} = A \quad \text{required}$$

$$C \rightarrow A \quad C^+ - \{C \rightarrow A\} = CDB \quad \text{required}$$

$$C \rightarrow D \quad C^+ - \{C \rightarrow D\} = \overbrace{CABD}^{\text{extra}}$$

$$F' = \{ \overset{\checkmark}{A \rightarrow D} \quad \overset{\checkmark}{C \rightarrow A} \quad \overset{\checkmark}{C \rightarrow B} \quad \underline{E \rightarrow A} \quad E \rightarrow D \}$$

$$E \rightarrow A \quad E^+ - \{E \rightarrow A\} = ED \quad \text{required}$$

$$E \rightarrow D \quad E^+ - \{E \rightarrow D\} = \overbrace{EAD}^{\text{extra}}$$

$$F' = \{A \rightarrow D \quad C \rightarrow A \quad C \rightarrow B \quad E \rightarrow A\}$$

$$F = \{ \overset{\checkmark}{A \rightarrow D} \quad \overset{\checkmark}{BC \rightarrow AD} \quad \overset{\checkmark}{C \rightarrow B} \quad \overset{\checkmark}{E \rightarrow AD} \}$$

$$BC \rightarrow AD$$

$$BC^+ \rightarrow BCDA \quad \text{proved}$$

$$E \rightarrow AD$$

$$E^+ \rightarrow \underline{EAD} \quad \text{proved}$$

Normalization Quiz

① $R(A\ B\ C\ D\ E\ F\ G\ H\ I\ J\ K\ L)$

$F \rightarrow G$

$H \rightarrow I\ J$

normalize till 3NF

$J \rightarrow K$

$L \rightarrow A\ H\ F$

①5

$B \rightarrow C\ D$

$L\ B \rightarrow E$

② Identify the normal form

$R(A\ B\ C\ D\ E\ F\ G\ H\ I)$

$AB \rightarrow C$

⑤

$BD \rightarrow EF$

$AD \rightarrow G\ H$

$A \rightarrow \underline{I}$

Revision

$R(W \times Y Z)$

$$Z \rightarrow W$$

$$Y \rightarrow XZ$$

$$XW \rightarrow Y$$

L	B	R
	W	
	X	
	Y	
	Z	

$$W^+ \rightarrow W$$

$$X^+ \rightarrow X$$

$$Y^+ \rightarrow Y X Z W \quad \checkmark$$

$$Z^+ \rightarrow Z W$$

$$PA = \{Y\}$$

$$\underline{XW^+} \rightarrow X W Y Z$$

	3NF	2NF
$\rightarrow \underline{Z} \rightarrow \underline{W}$	(X)	✓
$\underline{Y} \rightarrow X$	✓	✓
$\underline{Y} \rightarrow Z$	✓	✓
$\rightarrow \underline{XW} \rightarrow Y$	✓	(✓)
SK		
PA		
✓ AND X		
X		

$\alpha \rightarrow \beta$	3NF	2NF
SK	✓	
SK		

Let XYZ be a CK

$$PA = \{X, Y, Z\}$$

✓	X
✓	Y
✓	Z
	XY
	XZ
	YZ

